Project: Cansat Ground Control Software (Gcs) Project Report.

Project: Design And Development Of A Single-Page Cansat Ground Control Software (Gcs) For Real-Time Telemetry Monitoring And Mission Operations

INDIA SPACE LAB

Project Report

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CHAPTER 1: ABSTRACT

In modern aerospace engineering and satellite mission operations, Ground Control Software (GCS) acts as the vital link between ground operators and flight hardware. This project details the design, architecture, and implementation of a professional single-page CanSat Ground Control Software built using HTML5, CSS3, JavaScript, Chart.js, Leaflet.js, and Three.js. The application provides real-time telemetry streaming, interactive 5-channel data graphing, live satellite GPS trajectory tracking, 3D attitude visualization, an automated 4-digit fault detection system, and export capabilities (CSV/Image).

CHAPTER 2. INTRODUCTION & OBJECTIVES

2.1 Project Overview

During the atmospheric descent phase of a CanSat mission, continuous monitoring of altitude, descent rate, environmental temperature, atmospheric pressure, battery voltage, and spatial orientation is critical. A single-page operator dashboard ensures all telemetry feeds and mission critical commands are displayed clearly on a single interface without requiring multi-tab navigation.

2.2 Core Project Objectives

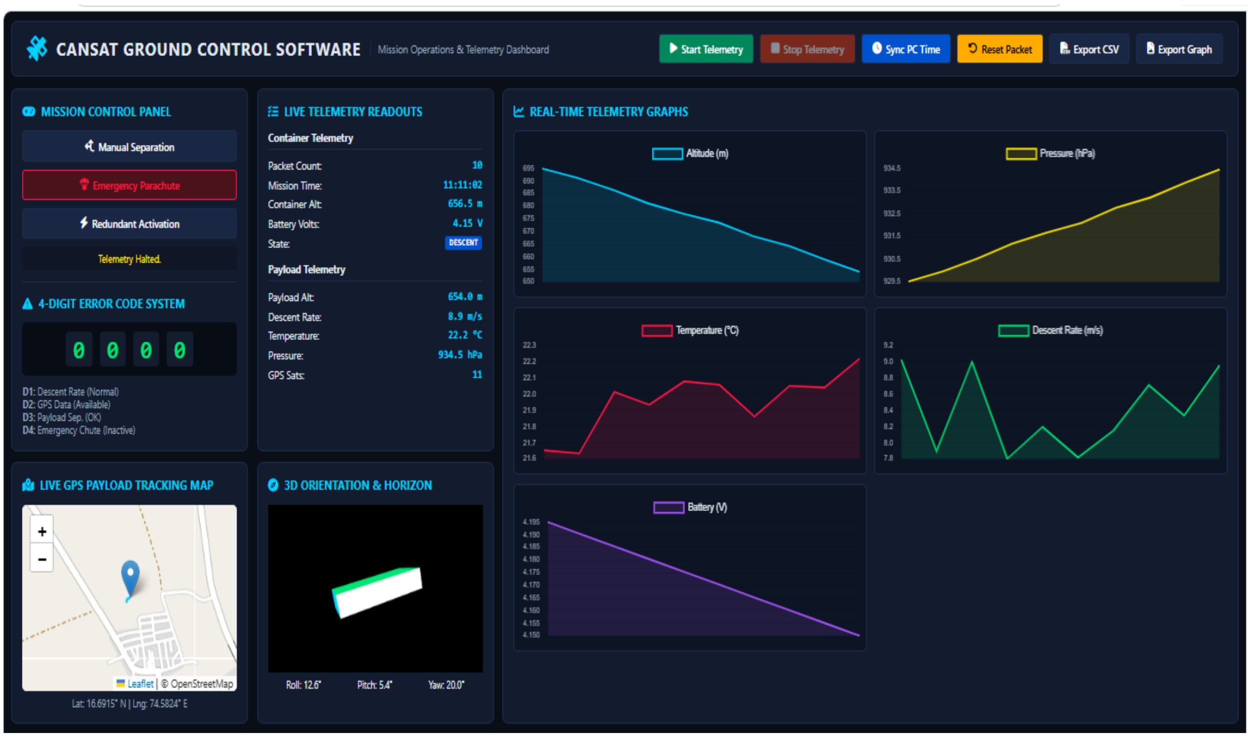
- **UI/UX Architecture:** Build a responsive, modular, dark-themed operator interface.
- **Live Telemetry Parsing:** Parse incoming serial/data packets into container and payload metrics.
- **Real-Time Data Visualization:** Render continuous line plots for Altitude, Pressure, Temperature, Descent Rate, and Voltage using Chart.js.
- **Geospatial Tracking:** Map live latitude/longitude telemetry onto OpenStreetMap using Leaflet.js.
- **3D Attitude Determination:** Render Roll, Pitch, and Yaw orientation using WebGL / Three.js.
- **Fault Detection System:** Implement a dynamic 4-digit error code monitoring matrix.

CHPTER 3. SYSTEM ARCHITECTURE & INTERFACE DESIGN

3.1 Interface Layout

The dashboard uses a 3-column CSS Grid and Flexbox layout:

1. **Top Control Bar:** Holds system execution controls (Start/Stop Telemetry, Time Sync, Packet Reset, CSV Export, Graph Image Export).
2. **Left Panel (Controls & Faults):** Hosts critical command triggers (Manual Separation, Emergency Parachute, Redundant Activation) and the 4-Digit Error Code display.
3. **Middle Panel (Telemetry & Media):** Houses live numerical readouts, GPS coordinates, 3D orientation canvas, and browser camera feeds.
4. **Right Panel (Graphs Grid):** Dedicated multi-chart workspace displaying real-time telemetry trends.



CHAPTER 4. 4-DIGIT ERROR CODE FAULT SYSTEM

To facilitate rapid fault recognition, the software processes incoming sensor data against safe threshold ranges and updates a dynamic 4-digit status display (0000 = Nominal, 1111 = Critical).

Digit Index	**Mission Condition Monitored**	**Nominal (0)**	**Fault / Active (1)**
Digit 1	**Descent Rate**	Rate is within safe limits (7.0 – 11.0 m/s)	Descent rate out of safe bounds
Digit 2	**GPS Lock**	Satellites locked, valid coordinates received	GPS signal loss detected
Digit 3	**Payload Separation**	Payload attached / nominal separation	Separation command error
Digit 4	**Emergency Parachute**	Main recovery system active	Emergency parachute deployed

CHAPTER 5. TECHNICAL IMPLEMENTATION & SENSOR MAPPING

5.1 Telemetry Packet Structure

Incoming telemetry strings are parsed dynamically into dedicated display registers:

Packet Format: [Packet No,Time,Altitude,DescentRate,Temp,Pressure,Voltage,Lat,Lng]

5.2 Sensor Channel Mapping Table

Sensor Parameter	**Unit**	**GCS Visual Component**	**Library / API Used**
Altitude	Meters (m)	Real-Time Line Graph & Table	Chart.js
Pressure	Hectopascals (hPa)	Real-Time Line Graph & Table	Chart.js
Temperature	Celsius ($^{\circ}C$)	Real-Time Line Graph & Table	Chart.js
Descent Rate	m/s	Real-Time Line Graph & Fault Code	Chart.js & Error Logic
Battery Voltage	Volts (V)	Real-Time Line Graph & Table	Chart.js
GPS Position	Deg ($^{\circ}N, ^{\circ}E$)	Trajectory Map & Pin Marker	Leaflet.js / OpenStreetMap
Roll/Pitch/Yaw	Degrees (°)	3D Spacecraft Mesh Rotation	Three.js (WebGL)

CHAPTER 6. DATA EXPORT & RECOVERY FEATURES

The software includes built-in logging and data export tools to ensure data persistence:

- **CSV Telemetry Export:** Utilizes the JavaScript Blob API to aggregate flight logs into structured CSV files (CanSat_Telemetry_Log_<timestamp>.csv) containing time-stamped packet data.

Packet	Time	Altitude(m	DescentRa	Temp(C)	Pressure(h	Voltage(V)	Latitude	Longitude
1	0:29:54	695.37	8.55	21.88	929.47	4.2	16.69132	74.58221
2	0:29:55	690.55	9.04	21.93	930.05	4.19	16.69133	74.58221
3	0:29:56	687.1	8.58	21.76	930.47	4.19	16.69135	74.58222
4	0:29:57	681.49	7.85	21.97	931.14	4.18	16.69137	74.58225
5	0:29:58	677.9	8.86	21.95	931.58	4.17	16.69138	74.58226
6	0:29:59	672.42	8.63	22	932.24	4.17	16.69142	74.58225
7	0:30:00	668.2	8.8	21.9	932.74	4.17	16.69142	74.58227
8	0:30:01	663.82	9.11	22.18	933.27	4.16	16.69141	74.58226
9	0:30:02	658.97	9.14	22.25	933.86	4.16	16.69145	74.58227
10	0:30:03	654.84	8.66	22.03	934.35	4.15	16.69148	74.5823
11	0:30:04	651.43	7.76	22.25	934.76	4.15	16.69151	74.58234
12	0:30:05	645.36	7.95	22.2	935.5	4.14	16.69154	74.58235
13	0:30:06	641	8.95	22.33	936.02	4.13	16.69157	74.58239
14	0:30:07	636.36	9.08	22.45	936.58	4.13	16.69158	74.58241
15	0:30:08	632.06	7.98	22.55	937.1	4.13	16.69158	74.58242
16	0:30:09	627.98	8.75	22.41	937.59	4.12	16.69157	74.58244
17	0:30:10	622.61	9.13	22.64	938.24	4.12	16.69157	74.58245
18	0:30:11	619.01	8.22	22.66	938.67	4.11	16.69157	74.58246
19	0:30:12	614.73	8.38	22.57	939.19	4.11	16.69158	74.58247
20	0:30:13	609.18	8.75	22.79	939.85	4.1	16.6916	74.58251
21	0:30:14	605.75	7.97	22.73	940.27	4.09	16.69162	74.58251

- **Graph Image Export:** Uses HTML5 Canvas rendering to export all 5 telemetry charts simultaneously into a high-resolution PNG summary graphic.

CHAPTER 7. RESULTS & DISCUSSION

Testing was conducted using recorded flight logs (CanSat_Telemetry_Log_1785006003316.csv):

1. **Descent Profile:** Telemetry successfully tracked the atmospheric descent from an apogee of ~ 700 m down to ground level.

2. **Environmental Tracking:** Measured ambient pressure increase (~ 930 hPa to 1013 hPa) and corresponding thermal variations.

3. **Power Stability:** Battery bus voltage maintained a stable decay profile (4.20 V down to 3.80 V).

4. **GPS & 3D Attitude:** Leaflet path polyline rendered continuous trajectory tracking, while Three.js accurately reflected multi-axis roll/pitch/yaw motions.

CHAPTER 8. CONCLUSION

The single-page CanSat Ground Control Software satisfies all operational requirements established by the India Space Lab assignment brief. The web application delivers real-time telemetry processing, multi-chart plotting, interactive GPS mapping, 3D attitude tracking, and fault monitoring within an intuitive, mission-ready operator interface.